PAPER_1761 — 21cm Dark Ages Temperature = T_21 = −D_phys × A_5 × β_i × 2 = −289.39 mK (vs EDGES −289)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Cosmology **Date:** June 18, 2026 **Status:** CLOSED — 0.14% closure (PAPER_132x-135x astrophysics + condensed matter)

Observation

PAPER_1333 (PAPER_132x-135x corpus) gives:

`T_21 = −D_phys × A_5 × β_i × 2 = −289.39 mK (vs EDGES −289)`

Status: **0.14%**.

UQFF Closed Identity

```
T_21 = −D_phys × A_5 × β_i × 2 = −289.39 mK (vs EDGES −289)    0.14%
```

The expression uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Standard frameworks address this differently. UQFF supplies a structural integer-primitive form.

Reference

- Source: PAPER_1333
- Related: PAPER_1746-1755 (tier-33); PAPER_1736-1745 (tier-32)
- Calculator dispatch: `calculate_paradox({"paradox": "t_21cm_minus_289_mk"})`

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